

TRAI asks telcos to use open source

The Telecom Regulatory Authority of India (TRAI) has asked mobile operators to adopt open standards, open source technology as well as open software code in order to offer unbiased Internet connections. According to Ram Sewak Sharma, TRAI's chairman, "No service can be hostage to a particular technology." He also added that by adopting open source (software), companies will benefit by avoiding vendor lock-in and offer unbiased Internet connectivity to everyone. He also stressed the need to use mobile phones as a means of digital connectivity and an instrument for digital transactions in India.

Digital connectivity will also reduce the pressure on the spectrum, and TRAI has sent its recommendations to the government regarding this. Regarding the issue of call drops, Sharma stated that in the recent round of drive tests, the regulator had obtained recommendations from the telecom operators and TRAI will be analysing these, prior to taking the appropriate steps. TRAI is also going to cover seven cities in the subsequent round of drive tests, which generally take place after every three months.



On the much talked about issue of differential pricing, which involved a review process taking place after two years, the chairman pointed out that two years is too long a time for a technology-driven industry. TRAI will take into account the policies that are still relevant and those that need to be upgraded or changed during that time frame.

Linux will power the future robots

Erle Robotics is a company making cost-effective components for DIY robotics projects as well as drones. Its offering, the PXFmini—a shield crafted for the US\$ 5 Raspberry Pi Zero—that comes with a suite of sensors which only a couple of years ago would have cost thousands of dollars, includes gravity, gyroscope, compass, pressure, temperature and battery sensors. The most noteworthy feature about it is the use of Linux, which has emerged as the most exciting robotics platform in use. The systems that are built with Linux vary from bots hacked together in home garages to the most sophisticated machines that come out of the fastest-growing hardware start-ups in the Bay Area.



Linux seems to be the future of robotics. This is mainly due to the proliferation of Raspberry Pi microcontrollers that are based on an ARM processor, which can run on Linux. The ARM architecture implies that certain parts of Linux have not been ported to the Raspberry Pi;

however, for basic robotics controls, the Raspberry Pi with Linux is the perfect fit for certain very capable robots and drones.

There's also something known as the Robotic Operating System (ROS), which is more an open source framework than an OS, released by Willow Garage under a free licence in 2009. It is now being maintained by the Open Source Robotics Foundation (OSRF), which announced in 2014 that it plans to add ARM Linux support to the ROS. Through an open source approach, ROS has compiled many of the advances in robotics controls and operation. This gives robotics scientists the ability to access precompiled packages, which permit them to make systems through common sensors including cameras and force sensors.

Owing to ROS and Linux, start-ups including Saviok and Fetch Robotics are coming out with new robots without needing to worry about motion planning and lower level controls. This saves many years of development and highlights the power of an open source ecosystem in terms of technological development.

Microsoft's Chakra goes open source

Over the past one year, it has been observed that many of Microsoft's offerings are being open sourced. As per a recent announcement by Microsoft, Chakra, the JavaScript engine powering Microsoft's latest browser, Edge, is going to be open sourced under the MIT license. The new open sourced version is going to be called ChakraCore.

Gaurav Seth, Chakra's principal project manager, said, "We've just made the sources for ChakraCore available under the MIT license at the ChakraCore GitHub repository. Going forward, we'll be developing the key components of Chakra in the open."

However, the ChakraCore will not be limited to the Edge browser. According to Seth, it powers Universal Windows applications through all form factors, including services like Azure DocumentDB, Cortana and Outlook.com. Using Windows 10, Microsoft has made it possible for Node.js to run with Chakra. This will assist it in advancing its reach and make it available on a new IoT platform, namely, Windows 10 IoT Core.

Microsoft is also planning to port ChakraCore to Linux.